

CHAPTER 9

CHOICE'S CONSEQUENCES

We will begin this chapter with a few results about countability whose proofs illustrate the difference between $\mathbf{AC}_{\mathbb{N}}$, $\mathbf{DC}$ and $\mathbf{AC}$, but our main task is to establish some important consequences of the full Axiom of Choice, including the basic laws of *cardinal arithmetic*. The telltale mark ($\mathbf{AC}$) will grace practically all the numbered propositions.

9.1. Theorem. *Every infinite set has a countable, infinite subset, and so for every cardinal number κ , either $\kappa <_c \aleph_0$ or $\aleph_0 \leq_c \kappa$.*

PROOF. If A is infinite, obviously

$$(\forall u \in A^*)(\exists y \in A)(\forall i < \text{lh}(u))[u(i) \neq y].$$

It follows from $\mathbf{DC}$ that there exists a sequence $f : \mathbb{N} \rightarrow A$ such that

$$(\forall n)(\forall i < n)[f(i) \neq f(n)],$$

and the image $f[\mathbb{N}]$ is a countable, infinite subset of A . The second assertion is trivial, taking cases on whether κ is finite or infinite. $\dashv$

The point of the second assertion of the theorem is that while the general property of Cardinal Comparability requires the full Axiom of Choice, the special (and significant) case of comparability with $\aleph_0$ is a theorem of $\mathbf{ZDC}$. In fact, it is possible to prove Theorem 9.1 using the Countable Principle of Choice $\mathbf{AC}_{\mathbb{N}}$ instead of $\mathbf{DC}$, but the proof is somewhat more technical, cf. Problem **x9.1**. This is a general fact about the relation between $\mathbf{DC}$ and $\mathbf{AC}_{\mathbb{N}}$: many results whose natural proofs call for $\mathbf{DC}$ follow from the weaker principle, with some additional effort.

Theorem 9.1 also settles the relation between infinite and Dedekind-infinite sets.

9.2. Corollary. *A set A is finite if and only if it is Dedekind-finite by 4.31, i.e., if there exists no injection $\pi : A \rightarrow B \subsetneq A$ from A into one of its proper subsets.*

PROOF. Finite sets are Dedekind-finite by the Pigeonhole Principle, 5.27. If A is infinite, let $f : \mathbb{N} \rightarrow A$ enumerate without repetitions some infinite, countable subset of it. The injection

$$\pi(x) = \begin{cases} f(n+1) & \text{if for some } n, x = f(n), \\ x & \text{if } x \notin f[\mathbb{N}] \end{cases}$$

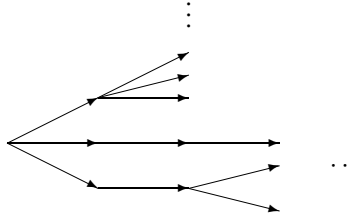


FIGURE 9.1. Non-empty tree.

witnesses that A is Dedekind-infinite, since $\pi[A] = A \setminus \{f(0)\}$. $\dashv$

Next we consider an elementary but very useful result about trees, whose proof offers an additional illustration of the use of **DC** and its relation to **AC_N**.

9.3. Definition. A **tree**²² on a set E is any set $T \subseteq E^*$ of strings from E which is closed under the relation of initial segment,

$$u \sqsubseteq v \in T \implies u \in T.$$

By (5-16), for strings, $u \sqsubseteq v \iff u \subseteq v$.

A lot of terms are used in the study of trees, most of them deriving from our picturing trees as, well, trees. The members of T are its **nodes** or **finite branches**, and every non-empty tree has $\emptyset$ as its least node, the **root**. If $u \star \langle x \rangle \in T$, then u is a **parent** of $u \star \langle x \rangle$ and $u \star \langle x \rangle$ is a **child** of u in T . Each node other than the root has exactly one parent, but may have many children; if it has none, it is a **terminal node** or **leaf**. With each node u we associate the **subtree**

$$T_u =_{\text{df}} \{w \in T \mid w \sqsubseteq u \vee u \sqsubseteq w\} \quad (9-1)$$

of nodes comparable with u . Easily,

$$T_u = \{w \mid w \sqsubseteq u\} \cup \bigcup \{T_v \mid v \text{ is a child of } u\}. \quad (9-2)$$

9.4. Exercise. Show (9-2).

The **infinite branches** of a tree are its infinite sequences, and we collect them in the **body** of T ,

$$[T] =_{\text{df}} \{f : \mathbb{N} \rightarrow E \mid (\forall n)[\overline{f}(n) \in T]\}. \quad (9-3)$$

Every infinite branch of a tree involves an infinite number of distinct nodes, so finite trees have empty bodies. It is also easy to construct infinite trees with empty bodies:

9.5. Exercise. Show that the tree

$$T = \{u \in \mathbb{N}^* \mid (\forall i < \text{lh}(u), i > 0)[u(i-1) > u(i)]\}$$

on the natural numbers is infinite but has no infinite branch.

²²Trees occur in many branches of mathematics, differently defined depending on the special needs of the field. The present definition is the most general we will need in these Notes.

9.6. Definition. A tree T is **finitely branching** if every node of T has at most finitely many children.

Notice that the tree in Exercise 9.5 is not finitely branching (at the root), and it could not be, by the following, basic result.

9.7. König's Lemma. *Every infinite, finitely branching tree has at least one infinite branch.*

PROOF. Suppose $T \subseteq E^*$ is infinite, finitely branching, and let

$$S =_{\text{df}} \{u \in T \mid T_u \text{ is infinite}\}.$$

This is the subtree of those nodes in T which are comparable with infinitely many nodes. Since $T_\emptyset = T$ is infinite by hypothesis, the root $\emptyset \in S$, and (9-2) implies that S has no leaves,

$$(\forall u \in S)(\exists x \in T)[u \star \{x\} \in S],$$

because each u has at most finitely many children and the infinite set S_u cannot be a finite union of finite sets. By the strong version of **DC** in Proposition 8.14, there exists some $f : \mathbb{N} \rightarrow E$ such that for every n , $\overline{f}(n) \in S$ and $f(n+1)$ is a child of $\overline{f}(n)$, so that f is an infinite branch of S —and hence of T . $\dashv$

König's Lemma is very useful, especially in the following, more “constructive” version.

9.8. Definition. A set of nodes $B \subseteq T$ is a **bar** in a tree T , if every infinite branch of T passes through at least one node of B ,

$$(\forall f \in [T])(\exists n)[\overline{f}(n) \in B].$$

9.9. Fan Theorem. *If T is a finitely branching tree and B is a bar for T , then there exists a finite subset*

$$B_0 = \{u_1, \dots, u_n\} \subseteq B$$

which is also a bar of T .

PROOF. Let B_0 comprise the minimal members of the bar B ,

$$B_0 =_{\text{df}} \{u \in B \mid (\forall v \subsetneq u)[v \notin B]\},$$

and notice that B_0 is also a bar, because if $f \in [T]$ and n is least such that $\overline{f}(n) \in B$, then $\overline{f}(n) \in B_0$. Let S be the tree of all initial segments of the nodes in B_0 ,

$$S =_{\text{df}} \{v \in T \mid (\exists u \in B_0)[v \subseteq u]\}.$$

Now S is a finitely branching tree (a subtree of T), and its leaves are precisely the nodes in B_0 , because no member of B_0 is a proper initial segment of another. Thus, S cannot have an infinite branch, since B_0 is a bar for T . By König's lemma then, S is finite, and so its subset B_0 is also finite. $\dashv$

The surprisingly simple proof of König's Lemma is typical of arguments from **DC**, partly because its basic structure calls for **DC**, but also because of the following two reasons:

(1) König's Lemma can be proved for every tree T on a well orderable set E with no use of choice principles, Problem **x9.3**. In many applications, $E = \mathbb{N}$ or E is finite, and then we need no choice hypotheses whatsoever.

(2) Like **9.1**, König's Lemma can be proved by appealing to **AC** $_{\mathbb{N}}$ rather than **DC**, Problem **x9.4**.

Many of the applications of the full Axiom of Choice have the following form: first we state and prove in **ZDC** (or even with no choice at all) some interesting proposition about well orderable sets, and then we infer the result we want for all sets by appealing to the Wellordering Theorem. Typical is the following generalization of the Hypothesis of Cardinal Comparability where (for the first and last time) we will state separately the corollary about all sets.

9.10. Theorem. Well foundedness of $\leq_c$. (1) *For every non-empty class $\mathcal{E}$ of well orderable sets, there exists some $A_0 \in \mathcal{E}$ such that for every $A \in \mathcal{E}$, $A_0 \leq_c A$.*

(2) **(AC)** *Every non-empty class $\mathcal{E}$ of sets has a $\leq_c$ -least member.*

PROOF. By **7.33**, let $U_0 = (A_0, \leq_0)$ be a $\leq_o$ -least well ordered set with field in $\mathcal{E}$. If $A \in \mathcal{E}$, then there exists some wellordering $\leq$ of A and by the choice of U_0 , $(A_0, \leq_0) \leq_o (A, \leq)$, so that, in particular, $A_0 \leq_c A$ since every initial similarity is an injection. $\dashv$

9.11. Lemma. The next cardinal. *For every well orderable cardinal number κ , the cardinal*

$$\kappa^+ =_{\text{df}} |\chi(\kappa)| \quad (9-4)$$

is also well orderable and it is least among the well orderable cardinals bigger than κ , i.e.,

$$\kappa <_c \kappa^+, \quad \kappa <_c \lambda \implies \kappa^+ \leq_c \lambda, \quad (9-5)$$

*for every well orderable cardinal λ . Here $\chi(\kappa)$ is the Hartogs well ordered set of κ , defined in **7.34**.*

PROOF. Since κ^+ is well orderable, it is comparable with κ , and it cannot be $\leq_c \kappa$ by Hartogs' Theorem **7.34**, so $\kappa <_c \kappa^+$. The minimality of $\chi(\kappa)$ implies the rest. $\dashv$

We set

$$\aleph_1 =_{\text{df}} \aleph_0^+, \quad \aleph_2 =_{\text{df}} \aleph_1^+, \dots \quad (9-6)$$

9.12. Exercise. **(AC)** *Since (with **AC**) every two cardinal numbers are comparable, the Continuum Hypothesis **CH** and the Generalized Continuum Hypothesis **GCH** can also be expressed by simple equations of cardinal arithmetic:*

$$\mathbf{CH} \iff 2^{\aleph_0} =_c \aleph_1, \quad \mathbf{GCH} \iff (\forall \kappa \geq_c \aleph_0)[2^\kappa =_c \kappa^+]. \quad (9-7)$$

This, unfortunately, does not help their resolution.

The next Lemma is often useful in arguments about well orderable sets.

9.13. Definition. A **best wellordering** of a set A is any wellordering $\leq$ of A in which every initial segment is smaller in cardinality than A ,

$$(\forall x \in A)[\text{seg}(x) <_c A].$$

9.14. Lemma. (1) *Every well orderable set admits a best wellordering.*

(2) *If $\leq_A, \leq_B$ are best wellorderings of A and B , then*

$$\text{if } A =_c B, \text{ then } (A, \leq_A) =_o (B, \leq_B).$$

In particular, any two best wellorderings of the same set are similar.

PROOF. (1) Let $U = (A, \leq)$ be $\leq_o$ -least in the class of all well ordered sets with field A , and suppose (towards a contradiction) that there exists some $x \in A$, $A \leq_c \mathbf{seg}_U(x)$. This yields an injection $\pi : A \rightarrow \mathbf{seg}_U(x)$ and the relation

$$u \leq' v \iff_{\text{df}} \pi(u) \leq \pi(v) \quad (u, v \in A)$$

is evidently a wellordering of A which is $\leq_o \mathbf{seg}_U(x)$ by 7.32, hence $<_o U$, contrary to the choice of U . (2) Suppose $U = (A, \leq_A)$, $V = (B, \leq_B)$ and (towards a contradiction) $U =_o \mathbf{seg}_V(x)$ for some $x \in B$. The similarity $\pi : A \rightarrow \mathbf{seg}_V(x)$ witnesses that $A =_c \mathbf{seg}_V(x) <_c B$, which is contrary to the hypothesis $A =_c B$. $\dashv$

Every best wellordering of a countable, infinite set is similar with the natural wellordering of $\mathbb{N}$, and we can use best wellorderings to show that many properties of countable sets hold for all well orderable sets. Typical is the next result, which generalizes the identity $\aleph_0^2 =_c \aleph_0$ and shows that the transfinite arithmetic of binary addition and multiplication is trivial.

9.15. Lemma. *For every infinite, well orderable set C , $C \times C =_c C$.*

PROOF. Assume the contrary towards a contradiction, let C be a $\leq_c$ -least counterexample by 9.10, and let $\leq$ be a best wellordering of C . By the choice of C , for every infinite point $x \in C$,

$$\begin{aligned} |\mathbf{seg}(x)| + |\mathbf{seg}(x)| &=_c 2 \cdot |\mathbf{seg}(x)| \\ &\leq_c |\mathbf{seg}(x)| \cdot |\mathbf{seg}(x)| =_c \mathbf{seg}(x) <_c C. \end{aligned} \quad (9-8)$$

The key step in the proof is the following definition of a new wellordering of the product $C \times C$, due to Gödel, which we have already met (somewhat concealed) in the proof of 5.32. We set

$$\begin{aligned} (x_1, y_1) \leq_g (x_2, y_2) &\iff_{\text{df}} [\max(x_1, y_1) < \max(x_2, y_2)] \\ &\vee [\max(x_1, y_1) = \max(x_2, y_2) \ \& \ x_1 < x_2] \\ &\vee [\max(x_1, y_1) = \max(x_2, y_2) \ \& \ x_1 = x_2 \\ &\quad \& \ y_1 \leq y_2]. \end{aligned} \quad (9-9)$$

The maxima here are obviously computed relative to the ordering $\leq$.

Sublemma. *The relation $\leq_g$ is a wellordering of $C \times C$.*

Proof. It is easy to verify that $\leq_g$ is a linear ordering. To show that it is a wellordering, suppose $X \subseteq C \times C$ is not empty: let w^* be $\leq$ -least such that for some $(x, y) \in X$, $\max(x, y) = w^*$; next let x^* be $\leq$ -least such that for some y , $(x^*, y) \in X$ and $\max(x^*, y) = w^*$; and finally let y^* be $\leq$ -least such that $(x^*, y^*) \in X$, $\max(x^*, y^*) = w^*$. It follows that (x^*, y^*) is the least member of X . $\dashv$ (Sublemma)

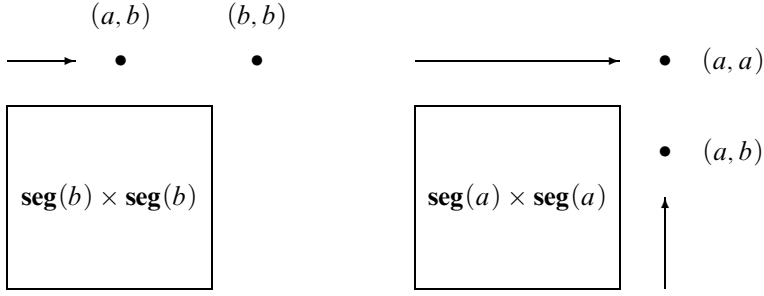


FIGURE 9.2. Initial segments of the Gödel wellordering.

The well ordered sets $(C, \leq)$ and $(C \times C, \leq_g)$ are $\leq_o$ -comparable and by the choice of $(C, \leq)$, $(C \times C, \leq_g) \leq_o (C, \leq)$ is not possible, so we must have $C <_o C \times C$; thus there exists some pair (a, b) of members of C such that

$$(C, \leq) =_o \mathbf{seg}_{C \times C}((a, b)) = \mathbf{seg}_g((a, b)),$$

and we will reach the desired contradiction if we can show that the initial segment $\mathbf{seg}_g((a, b)) <_c C$. We consider the possibilities arising from the relative positions of a and b in $\leq$, and we use the fact that the point $\max(a, b)$ must be infinite.

CASE 1, $a = b$. From the definition of the Gödel wellordering,

$$(u, v) <_g (a, a) \iff [u < a \ \& \ v < a] \vee [u < a \ \& \ v = a] \vee [u = a \ \& \ v < a],$$

so that

$$\mathbf{seg}_g((a, a)) = (\mathbf{seg}(a) \times \mathbf{seg}(a)) \cup (\mathbf{seg}(a) \times \{a\}) \cup (\{a\} \times \mathbf{seg}(a)),$$

and by repeated applications of (9-8),

$$|\mathbf{seg}_g((a, a))| \leq_c |\mathbf{seg}(a)|^2 + |\mathbf{seg}(a)| \cdot 2 \leq_c |\mathbf{seg}(a)| \cdot 3 <_c C.$$

CASE 2, $a < b$. Now $\max(a, b) = b$,

$$(u, v) <_g (a, b) \iff [u < b \ \& \ v < b] \vee [u < a \ \& \ v = b],$$

so $\mathbf{seg}((a, b)) = (\mathbf{seg}(b) \times \mathbf{seg}(b)) \cup (\mathbf{seg}(a) \times \{b\})$ and a similar computation shows again that $|\mathbf{seg}_g((a, b))| <_c C$.

CASE 3, $a > b$. This time

$$(u, v) <_g (a, b) \iff [u < a \ \& \ v < a] \vee [u < a \ \& \ v = a] \vee [u = a \ \& \ v < b],$$

from which we reach a contradiction as in the preceding cases. $\dashv$

9.16. Theorem (The absorption laws). *If κ and λ are non-zero, well orderable cardinals and at least one of them is infinite, then*

$$\kappa + \lambda =_c \kappa \cdot \lambda =_c \max(\kappa, \lambda).$$

PROOF. Assuming that $0 <_c \kappa \leq_c \lambda$ and using the result $\lambda \cdot \lambda =_c \lambda$ from the Lemma, we compute

$$\lambda \leq_c \kappa + \lambda \leq_c \kappa \cdot \lambda \leq_c \lambda \cdot \lambda =_c \lambda. \quad \dashv$$

9.17. Corollary. (AC) For every indexed family of sets $(i \mapsto \kappa_i)_{i \in I}$ and every infinite κ , if $|I| \leq_c \kappa$ and for each $i \in I$, $\kappa_i \leq_c \kappa$, then $\sum_{i \in I} \kappa_i \leq_c \kappa$.

PROOF. Using AC and the hypothesis, choose for each $i \in I$ some injection $\pi_i : \kappa_i \rightarrow \kappa$, so that the mapping $((i, x) \mapsto (i, \pi_i(x)))$ is an injection of $\{(i, x) \mid i \in I \text{ \& } x \in \kappa_i\}$ into $I \times \kappa$. Thus

$$\sum_{i \in I} \kappa_i =_c |\{(i, x) \mid i \in I \text{ \& } x \in \kappa_i\}| \leq_c |I \times \kappa| =_c |I| \cdot |\kappa| =_c \kappa. \quad \dashv$$

To find interesting problems and results in cardinal arithmetic we must consider operations with infinitely many arguments, of which the simplest are the following.

9.18. Cardinal Minimum Lemma. There is a definite operation $\inf_c(\mathcal{E})$, such that for each non-empty family $\mathcal{E}$ of well orderable sets, the value $\kappa = \inf_c(\mathcal{E})$ has the following properties:

- (1) κ is a well orderable cardinal number.
- (2) For some $A \in \mathcal{E}$, $\kappa =_c A$.
- (3) For every $B \in \mathcal{E}$, $\kappa \leq_c B$.

In addition, these conditions determine the value $\inf_c(\mathcal{E})$ up to $=_c$, i.e., if κ is any set which satisfies (1) – (3), then $\kappa =_c \inf_c(\mathcal{E})$.

PROOF. If the cardinal assignment $|X|$ is strong by 4.21, then 9.10 implies that there exists exactly one cardinal number which satisfies the condition

$$\text{Least}(\mathcal{E}, \kappa) \iff (\exists A \in \mathcal{E})[(\forall B \in \mathcal{E})[A \leq_c B] \text{ \& } \kappa = |A|],$$

and we can set

$$\inf_c(\mathcal{E}) = \min(\mathcal{E}) = \text{the unique } \kappa \text{ such that } \text{Least}(\mathcal{E}, \kappa).$$

We need to do more, since we have only assumed that $|X|$ is a weak cardinal assignment and there may well exist many values of κ which satisfy $\text{Least}(\mathcal{E}, \kappa)$.

By the Lemma in the proof of Hartogs' Theorem 7.34, if $A \subseteq \bigcup \mathcal{E}$ and $\leq$ is a wellordering of A , then the well ordered set $U = (A, \leq)$ is similar with some proper initial segment of $W = \chi(\bigcup \mathcal{E})$, and hence every $A \in \mathcal{E}$ is equinumerous with some proper initial segment of W . Thus we can set

$$w =_{\text{df}} \text{the least } x \in W \text{ such that } (\exists A \in \mathcal{E})[A =_c \text{seg}_W(x)],$$

$$\inf_c(\mathcal{E}) =_{\text{df}} |\text{seg}_W(w)|.$$

Verification of the required properties of $\inf_c(\mathcal{E})$ is quite easy. $\dashv$

9.19. Exercise. Show the part of the theorem which follows the “in addition”.

9.20. Cardinal Supremum Lemma. There is a definite operation $\sup_c(\mathcal{E})$, such that for every non-empty family $\mathcal{E}$ of well orderable sets, the value $\kappa = \sup_c(\mathcal{E})$ has the following properties:

- (1) κ is a well orderable cardinal.
- (2) For every $A \in \mathcal{E}$, $A \leq_c \kappa$.
- (3) If B is well orderable and for all $A \in \mathcal{E}$, $A \leq_c B$, then $\kappa \leq_c B$.

In addition, these conditions determine the value $\sup_c(\mathcal{E})$ up to $=_c$, i.e. if κ is any set which satisfies (1) – (3), then $\kappa =_c \sup_c(\mathcal{E})$.

PROOF. Let $C = h(\bigcup \mathcal{E})$ be the Hartogs set for the union of $\mathcal{E}$, which by Hartogs' Theorem 7.34 is well orderable and greater in cardinality than every well orderable subset of $\bigcup \mathcal{E}$, including every $A \in \mathcal{E}$. We set

$$\sup_c(\mathcal{E}) =_{\text{df}} \inf_c(\{B \subseteq C \mid (\forall A \in \mathcal{E})[A \leq_c B]\})$$

and verify easily the conclusion of the Lemma. ⊢

Infinite sums and products were defined in 4.21. We cannot say much about these, because infinite sums are as trivial as the finite ones (Problem x9.15), and infinite products are as complex as the Generalized Continuum Hypothesis, because

$$2^\kappa =_c \prod_{i \in \kappa} 2.$$

There is, however, a very interesting inequality relating the two.

9.21. König's Theorem. (AC) For any two families of sets $(i \mapsto A_i)$ and $(i \mapsto B_i)$ on the same index set $I \neq \emptyset$,

$$\text{if } (\forall i \in I)[A_i <_c B_i], \text{ then } \bigcup_{i \in I} A_i <_c \prod_{i \in I} B_i. \quad (9-10)$$

In particular, for families of cardinals, $(i \mapsto \kappa_i)$ and $(i \mapsto \lambda_i)$,

$$\text{if } (\forall i \in I)[\kappa_i <_c \lambda_i], \text{ then } \sum_{i \in I} \kappa_i <_c \prod_{i \in I} \lambda_i. \quad (9-11)$$

PROOF. The hypothesis and AC yield for each i an injection $\pi_i : A_i \rightarrow B_i$; and since π_i cannot be a bijection, there also exists a function $c : I \rightarrow \bigcup_{i \in I} B_i$ such that for each i , $c(i) \in B_i \setminus \pi_i[A_i]$. We set

$$f(x, i) = \begin{cases} \pi_i(x), & \text{if } x \in A_i, \\ c(i), & \text{if } x \notin A_i, \end{cases}$$

$$g(x) = (i \mapsto f(x, i)).$$

If $x \neq y$ and x, y belong to the same A_i for some i , then

$$g(x)(i) = \pi_i(x) \neq \pi_i(y) = g(y)(i),$$

because π_i is an injection, and hence $g(x) \neq g(y)$. If no A_i contains both x and y , suppose $x \in A_i$, $y \notin A_i$; it follows that $g(x)(i) = \pi_i(x) \in \pi_i[A_i]$ and $g(y)(i) = c(i) \in B_i \setminus \pi_i[A_i]$ so that again $g(x) \neq g(y)$. We conclude that the mapping $g : \bigcup_{i \in I} A_i \rightarrow \prod_{i \in I} B_i$ is an injection, and hence

$$\bigcup_{i \in I} A_i \leq_c \prod_{i \in I} B_i.$$

Suppose, towards a contradiction that there existed a correspondence

$$h : \bigcup_{i \in I} A_i \twoheadrightarrow \prod_{i \in I} B_i,$$

so that these two sets are equinumerous. For every i , the function

$$h_i(x) =_{\text{df}} h(x)(i) \quad (x \in A_i)$$

is (easily) a function of A_i into B_i and by the hypothesis it cannot be a surjection; hence by **AC** there exists a function ε which selects in each B_i some element not in the image, i.e.,

$$\varepsilon(i) \in B_i \setminus h_i[A_i], \quad (i \in I).$$

By its definition, $\varepsilon \in \prod_{i \in I} B_i$, so there must exist some $x \in A_j$, for some j , such that $h(x) = \varepsilon$; this yields

$$\varepsilon(j) = h(x)(j) = h_j(x) \in h_j[A_j],$$

contrary to the characteristic property of ε .

The cardinal version (9-11) follows by applying (9-10) to $A_i = \{i\} \times \kappa_i$ and $B_i = \lambda_i$. $\dashv$

9.22. Exercise. (**AC**) *König's Theorem applies to the case $I = \kappa$, $A_i = \{i\}$ and $B_i = 2$ and yields*

$$\kappa = \bigcup_{i \in \kappa} \{i\} <_c \prod_{i \in \kappa} 2 =_c 2^\kappa,$$

i.e., the theorem of Cantor.

Despite its simplicity, König's Theorem implies immediately a non-trivial inequality about the cardinal number $\mathfrak{c}$ of the continuum which goes beyond Cantor's $\aleph_0 <_c \mathfrak{c}$. It is most naturally expressed using *cofinalities*.

9.23. Definition. The **cofinality** of a well orderable, infinite cardinal number κ is the least well orderable cardinal λ such that κ is the union of λ -many sets with cardinality smaller than κ . Precisely:

$$\text{cf}(\kappa) =_{\text{df}} \inf_c (\{I \subseteq \kappa \mid \text{for some indexed family } (i \mapsto K_i)_{i \in I}, \\ (\forall i \in I)[K_i <_c \kappa] \ \& \ \kappa = \bigcup_{i \in I} K_i\}).$$

Notice that the family of well orderable index sets whose $\inf_c$ we take is not empty; it contains κ , since

$$\kappa = \bigcup_{i \in \kappa} \{i\}. \quad (9-12)$$

The general properties of $\inf_c$ imply the following basic properties of the cofinality operation:

- (1) $\text{cf}(\kappa) \leq_c \kappa$.
- (2) $\kappa = \bigcup_{i \in \text{cf}(\kappa)} K_i$ for some indexed family $(i \mapsto K_i)$ such that $(\forall i \in \text{cf}(\kappa))[K_i <_c \kappa]$.
- (3) If λ is well orderable, $(\forall i \in \lambda)[L_i <_c \kappa]$ and $\kappa = \bigcup_{i \in \lambda} L_i$, then $\text{cf}(\kappa) \leq_c \lambda$.

Moreover, these conditions characterize $\text{cf}(\kappa)$ up to $=_c$.

A well orderable, infinite cardinal κ is **regular** if $\text{cf}(\kappa) =_c \kappa$, otherwise it is **singular**. It is convenient to define the operation $\text{cf}(\kappa)$ and the regularity condition for infinite, well orderable κ without assuming the full Axiom of Choice, but most results about these notions require **AC**.

9.24. Exercise. $\aleph_0$ is regular, because every finite sum of finite cardinals is finite.

9.25. Corollary. (AC) For each infinite cardinal number κ ,

$$\text{cf}(2^\kappa) > \kappa,$$

and in particular, $\text{cf}(\mathfrak{c}) > \aleph_0$, i.e., the continuum $\mathfrak{c}$ is not the union of countably many sets of cardinality $< \mathfrak{c}$.

PROOF. By König's Theorem, if $K_i < {}_c 2^\kappa$ for every $i \in \lambda$ with $\lambda \leq_c \kappa$, then

$$\bigcup_{i \in \lambda} K_i < {}_c \prod_{i \in \lambda} 2^\kappa = {}_c (2^\kappa)^\lambda = {}_c 2^{\kappa \cdot \lambda} = {}_c 2^\kappa,$$

which contradicts $\text{cf}(2^\kappa) \leq_c \kappa$. ⊥

9.26. Gödel's model L of the constructible sets satisfies the Generalized Continuum Hypothesis, so for each κ , $2^\kappa = {}_c \kappa^+$ is regular by Problem **x9.19**. Using Cohen's forcing method, it is possible to construct models of Zermelo's theory **ZDC** + **AC** in which $\mathfrak{c}$ is singular, with cofinality $\text{cf}(\mathfrak{c})$ some regular cardinal between $\aleph_0$ and $\mathfrak{c}$, for example $\aleph_1$.

We have left the basic properties of cofinalities for the problems, as they are very simple. We should remark, however, that it is not possible to study the topic seriously now, because without the Axiom of Replacement, it is not even possible to prove that singular cardinals *exist*!

Problems for Chapter 9

***x9.1.** Show Theorem **9.1** using only the constructive axioms (I) – (VI) and the Countable Principle of Choice **AC_N**.

x9.2. Consider a system of airline routes which connects the (possibly infinitely many) cities of some world and assume the following. (1) From each city, there are only finitely many cities to which one can fly non-stop. (2) It is possible to travel by air from every city to every other city. (3) It is not possible to keep flying forever without visiting the same Airport twice. Show that this world has only finitely many cities.

x9.3. Show König's Lemma **9.7** for the case where T is a tree on a well orderable set E , with no appeal to choice principles.

***x9.4.** Show König's Lemma **9.7** using only the constructive axioms (I) – (VI) and the Countable Principle of Choice **AC_N**.

x9.5. Suppose T is a finitely splitting tree and B is a bar for T . Show that there exists some integer k , such that for all infinite branches $f \in [T]$, some $\bar{f}(i) \in B$, with $i \leq k$.

***x9.6.** Suppose C is a well orderable set, $f : C \times C \rightarrow C$, $A \subseteq C$, and let

$$A_f =_{\text{df}} \bigcap \{X \subseteq C \mid A \subseteq X \text{ \& } f[X \times X] \subseteq X\}$$

be the **closure** of A under f . Define the sets $\{A_n\}_{n \in \mathbb{N}}$ by the recursion

$$A_0 = A, \quad A_{n+1} = A_n \cup f[A_n \times A_n],$$

and show that

$$A_f = \bigcup_{n \in \mathbb{N}} A_n.$$

Show also that if A is infinite, then $A_f =_c A$.

x9.7. Show that if C is well orderable, then so is the set C^* of all words (finite sequences) from C .

x9.8. If you used **AC**_ℕ or **DC** in Problems **x9.6** and **x9.7**, do them again, using no choice principles at all.

x9.9. Every Hartogs set $h(A)$ is best wellordered by $\leq_{\chi(A)}$.

x9.10. If $(n \mapsto \kappa_n)_{n \in \mathbb{N}}$ and $(n \mapsto \lambda_n)_{n \in \mathbb{N}}$ are sequences of cardinal numbers, and for every n , $\kappa_n \leq_c \lambda_n$, then

$$\sum_{n \in \mathbb{N}} \kappa_n \leq_c \sum_{n \in \mathbb{N}} \lambda_n, \quad \prod_{n \in \mathbb{N}} \kappa_n \leq_c \prod_{n \in \mathbb{N}} \lambda_n.$$

x9.11. (AC) If $(i \mapsto \kappa_i)_{i \in I}$ and $(i \mapsto \lambda_i)_{i \in I}$ are families of cardinal numbers on the same index set I and for all $i \in I$, $\kappa_i \leq_c \lambda_i$, then

$$\sum_{i \in I} \kappa_i \leq_c \sum_{i \in I} \lambda_i, \quad \prod_{i \in I} \kappa_i \leq_c \prod_{i \in I} \lambda_i.$$

x9.12. (AC) For every indexed family of sets $(i \mapsto A_i)_{i \in I}$,

$$|\prod_{i \in I} A_i| =_c \prod_{i \in I} |A_i|$$

and the same for sums, with “disjoint union” on the left.

x9.13. Explain the notation and prove the identity

$$\prod_{i \in I} \prod_{j \in J(i)} \kappa_{ij} =_c \prod_{\{(i,j) \mid i \in I \text{ \& } j \in J(i)\}} \kappa_{ij}.$$

x9.14. (AC) Prove the characterization of $\sup_c(\mathcal{E})$ claimed in **9.20**.

x9.15. (AC) For every family of infinite cardinal numbers $(i \mapsto \kappa_i)$ on a non-empty index set I ,

$$\sum_{i \in I} \kappa_i =_c \max(|I|, \sup_c(\{\kappa_i \mid i \in I\})).$$

x9.16. Prove that for every infinite, well orderable cardinal κ ,

$$\begin{aligned} \text{cf}(\kappa) = \inf_c \Big(\{ I \subseteq \kappa \mid \text{for some indexed family } (i \mapsto K'_i)_{i \in I}, \\ (\forall i, j \in I)[i \neq j \implies K'_i \cap K'_j = \emptyset] \text{ \& } (\forall i \in I)[K'_i <_c \kappa] \\ \text{\& } \kappa = \bigcup_{i \in I} K'_i \} \Big). \end{aligned}$$

***x9.17. (AC)** Prove that for every infinite cardinal κ ,

$$\text{cf}(\kappa) = \inf_c \left(\left\{ I \subseteq \kappa \mid \text{for some indexed family of cardinals } (i \mapsto \kappa_i)_{i \in I}, \right. \right. \\ \left. \left. (\forall i \in I)[\kappa_i <_c \kappa] \& \kappa =_c \sum_{i \in I} \kappa_i \right\} \right).$$

HINT. It helps to use Problem **x9.16**.

***x9.18.** Show that for every infinite cardinal κ , $\text{cf}(\text{cf}(\kappa)) =_c \text{cf}(\kappa)$, and hence $\text{cf}(\kappa)$ is always a regular cardinal

x9.19. (AC) For each infinite cardinal κ , the next cardinal κ^+ is regular. HINT. Problem **x9.17** simplifies the proof.

x9.20. (AC) Show that for each infinite cardinal κ , $\kappa <_c \kappa^{\text{cf}(\kappa)}$.

***x9.21. (AC)** Every partial ordering $\leq$ on a set P has a **linearization**, i.e., some linear ordering $\leq'$ of P exists such that $x \leq y \implies x \leq' y$.

The next problem gives the basic fact which relates inductive and directed-complete posets. Notice that by “chain” in a poset P we mean any subset $C \subseteq P$ which is linearly ordered by the poset partial ordering $\leq_P$; C is a *well ordered chain* if in addition, the restriction of $\leq_P$ to C is a wellordering. When we say that “ S is well orderable” for some $S \subseteq P$, we mean that S *admits some wellordering* $\leq$, which may be (and typically is) totally unrelated with the given partial ordering $\leq_P$ of P .

***x9.22.** If every well ordered chain in a poset $(P, \leq_P)$ has a least upper bound, then for every well orderable, directed subset S of P there exists a well ordered chain C with the following two properties.

(1) S is dominated by C , i.e., for each $x \in S$ there exists some $y \in C$ such that $x \leq_P y$.

(2) For each $y \in C$, there exists a directed subset $C_y \subseteq S$ such that $|C_y| <_c |S|$ and $y = \sup C_y$.

Notice that C may satisfy these conditions without being a subset of S . HINT (W. Allen). Towards a contradiction, let S be well orderable, directed and $\leq_c$ -least counterexample to the conclusion, verify first that S must be uncountable, and let $\leq$ be a best wellordering of S . Define the function $f : S \times S \rightarrow S$ so that $x, y \in S \implies x, y \leq_P f(x, y)$, and for every $x \in S$, set

$$C_x =_{\text{df}} \text{seg}(x)_f,$$

with the notation of Problem **x9.6**. Show that this is directed, that $\sup C_x$ exists for each $x \in S$, and that

$$C =_{\text{df}} \{\sup C_x \mid x \in S\}$$

is a well ordered chain in P which has properties (1) and (2) for S .

***x9.23. (AC)** The following three conditions are equivalent, for every poset P :

1. Every directed set in P has a least upper bound.
2. Every chain in P has a least upper bound.

3. Every well ordered chain in P has a least upper bound.

In particular: (AC) *A poset is inductive if and only if it is directed-complete, a dcpo.*

***x9.24.** (AC) Show that a monotone mapping $\pi : P \rightarrow Q$ on one inductive poset to another satisfies the identity

$$\pi(\sup S) = \sup \pi[S] \quad (9-13)$$

for every non-empty chain $S \subseteq P$, if and only if it satisfies (9-13) for every non-empty directed $S \subseteq P$.

x9.25. Prove that the characterization of continuity for mappings of the form $\pi : (A \rightarrow E) \rightarrow (B \rightarrow M)$ in **x6.24** holds for all sets A, E, B, M .

x9.26. (AC) **Finite Basis Lemma.** Let $\mathcal{S}$ be a non-empty family of subsets of some set V , such that

$$X \in \mathcal{S} \iff (\forall Y \subseteq X)[Y \text{ finite} \implies Y \in \mathcal{S}].$$

Show that $\mathcal{S}$ has a maximal member (under $\subseteq$).

***x9.27.** Let $\mathcal{S}$ be a family with the finite basis property as in **x9.26** and assume in addition that V is well orderable; show (without AC) that $\mathcal{S}$ has a maximal member.

x9.28. (AC) If you know what *vector spaces* are and the basic facts about *linear independence*, prove that every vector space has a basis. Prove also without AC, that every well orderable vector space has a basis. HINT. Apply **x9.26** or **x9.27** to the family of all linearly independent subsets of the given space.

***x9.29.** (AC) If you know something about fields and algebraic extensions, prove that every field has an algebraic closure. HINT. The usual argument for this runs as follows. We consider the class

$$\mathcal{A} =_{\text{df}} \{F \mid F \text{ is an algebraic extension of } K\} \quad (9-14)$$

partially ordered by

$$F_1 \subseteq F_2 \iff_{\text{df}} F_1 \text{ is a subfield of } F_2,$$

we notice that it is an inductive poset, so that it has a maximal element $\overline{K}$, and we verify that this $\overline{K}$ is algebraically closed. The argument is defective, because the class $\mathcal{A}$ in (9-14) is not a set. To correct it, in the interesting case where K is infinite, we need to notice that *every algebraic extension of K is isomorphic with some field $F =_c K$* , so we can replace $\mathcal{A}$ in (9-14) by

$$\mathcal{A}' =_{\text{df}} \{F \subseteq E \mid F \text{ is an algebraic extension of } K\}, \quad (9-15)$$

where E is some superset of K with cardinality greater than K .

***x9.30.** Prove that every well orderable field (and in particular, every countable field) has an algebraic closure. **HINT.** The idea is to avoid **AC** by using Transfinite Recursion to construct the closure explicitly. You still need the trick suggested in the previous problem. Do the countable case first, it clarifies which algebraic results are needed.